

Course 6044D

Semester VI

Theory

4 Credits

Healthcare Electronics

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044D as Healthcare Electronics in Semester VI for Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on emerging healthcare technology curricula and Internet of Medical Things (IoMT) frameworks. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Healthcare designs, wearable calibrations, telehealth configurations and data privacy policies must be based on approved engineering plans, certified medical hardware data, current healthcare standards and competent professional review.

Verified source note: Verified sources support wearables, RPM, IoMT, and digital health trends. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Healthcare Electronics studies the modern electronic systems, wearable devices and digital technologies used to monitor, diagnose and improve human health. It develops the ability to

analyze wearable sensor networks, evaluate remote patient monitoring (RPM) and telehealth systems, understand the Internet of Medical Things (IoMT), and coordinate digital health solutions while respecting the technical, clinical and privacy boundaries of electronic and healthcare engineering.

Malayalam support: ു handbook ുുുുുുുുുുു syllabus topic ുുുുു ുുുുുുുുു; ുുുുുുുു principle, diagram/procedure, application, common mistake ുുുുുു ുുുുുുുുുു.

Prerequisite foundation

Basic electronics, sensors and transducers, microcontrollers, and basic communication systems.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of healthcare electronics, wearables and digital health trends.
2. Analyze the design and operation of wearable sensors and health monitoring systems.
3. Evaluate the systematic integration of IoT in healthcare (IoMT) and telehealth platforms.
4. Explain the regulatory requirements, data privacy and future innovations in healthcare electronics.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO ഉള്ളത് exam-ന് ഉപയോഗിക്കേണ്ടതാണ്. Define, explain, apply എന്നിവ action words ഉപയോഗിക്കേണ്ടതാണ്.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals of healthcare electronics and wearable technologies.	Understand
CO2	Analyze the operation of wearable sensors and remote monitoring systems.	Apply
CO3	Evaluate IoMT architectures and telehealth systems for healthcare delivery.	Apply
CO4	Explain regulatory standards, data privacy and future trends in digital health.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Introduction to Healthcare Electronics and Wearables	Evolution of healthcare technology; Wearable electronics: Smartwatches, Fitness trackers, Smart patches; Flexible and Stretchable electronics; Market trends and impact.	Approx. 14 hours	CO1	Module 1
2	Wearable Sensors and Monitoring Systems	Wearable sensors: PPG (Photoplethysmography), Bio-impedance, Temperature, Motion (IMU); Signal acquisition and processing; Remote Patient Monitoring (RPM) architecture.	Approx. 16 hours	CO2	Module 2
3	IoMT and Telehealth Systems	Internet of Medical Things (IoMT): Architecture and connectivity; Telehealth and Telemedicine; Smart hospitals; Cloud-based health records; Interoperability standards (HL7, FHIR).	Approx. 16 hours	CO3	Module 3
4	Regulation, Privacy and Future Innovations	Medical device regulations: FDA, CE, CDSCO; Data privacy and security: HIPAA, GDPR; Ethics in healthcare AI; Future trends: Bio-integrated electronics, Digital twins.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Introduction to Healthcare Electronics and Wearables

Evolution of healthcare technology; Wearable electronics: Smartwatches, Fitness trackers, Smart patches; Flexible and Stretchable electronics; Market trends and impact.

CO1 · Approx. 14 hours

Wearable Sensors and Monitoring Systems

Wearable sensors: PPG (Photoplethysmography), Bio-impedance, Temperature, Motion (IMU); Signal acquisition and processing; Remote Patient Monitoring (RPM) architecture.

CO2 · Approx. 16 hours

IoMT and Telehealth Systems

Internet of Medical Things (IoMT): Architecture and connectivity; Telehealth and Telemedicine; Smart hospitals; Cloud-based health records; Interoperability standards (HL7, FHIR).

CO3 · Approx. 16 hours

Regulation, Privacy and Future Innovations

Medical device regulations: FDA, CE, CDSCO; Data privacy and security: HIPAA, GDPR; Ethics in healthcare AI; Future trends: Bio-integrated electronics, Digital twins.

CO4 · Approx. 14 hours

MODULE 1

Introduction to Healthcare Electronics and Wearables

Evolution of healthcare technology; Wearable electronics: Smartwatches, Fitness trackers, Smart patches; Flexible and Stretchable electronics; Market trends and impact.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Introduction to Healthcare Electronics and Wearables: identify the system, state its principle, follow the correct method and verify the result safely.

1. Healthcare Technology Evolution–

Healthcare electronics has evolved from bulky hospital equipment to portable and wearable devices. Digitization and connectivity enable continuous monitoring outside clinical settings. Key drivers include aging populations, rising healthcare costs, and the need for preventive care.

Study and application method

List three 'Emerging Trends' in healthcare electronics; explain the benefit of 'Wearable Technology' for chronic disease management.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three 'Emerging Trends' in healthcare electronics; explain the benefit of 'Wearable Technology' for chronic disease management.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിട്ടുണ്ട്. ഏതു diagram / circuit / formula / procedure നൽകിയിട്ടുണ്ട്. ഏതു result നൽകിയിട്ടുണ്ട് application നൽകിയിട്ടുണ്ട്. Technical terms English-ൽ എഴുതിക്കുക.

Common mistake / precaution: Wearable data is for monitoring, not self-diagnosis. Do not interpret wearable readings for clinical decisions without authorised medical consultation.

2. Flexible and Wearable Electronics–

Wearable devices require flexible substrates and low-power components. Smart patches use thin-film sensors for continuous glucose or ECG monitoring. Flexible batteries and antennas are essential for comfort and performance.

Study and application method

Describe the concept of 'Flexible Electronics' conceptually; list two types of 'Smart Wearables' used in healthcare.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the concept of 'Flexible Electronics' conceptually; list two types of 'Smart Wearables' used in healthcare.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ diagram / circuit / formula / procedure ഈ result application ഈ technical terms English-ൽ എങ്ങനെ വിശദീകരിക്കണം.

Common mistake / precaution: Skin irritation and biocompatibility are major concerns. All wearable designs must follow authorised material safety and dermatological standards.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Wearable Sensors and Monitoring Systems

Wearable sensors: PPG (Photoplethysmography), Bio-impedance, Temperature, Motion (IMU); Signal acquisition and processing; Remote Patient Monitoring (RPM) architecture.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

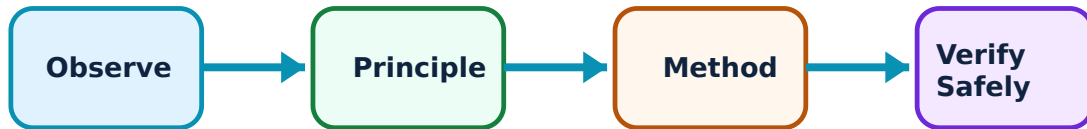
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wearable Sensors and Monitoring Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Physiological Sensing in Wearables–

PPG sensors use light to measure heart rate and SpO2. Bio-impedance sensors measure body composition and hydration. IMUs (Inertial Measurement Units) track activity levels and detect falls. These sensors must be low-power and highly integrated.

Study and application method

Explain the principle of 'PPG' for heart rate measurement; describe how 'Motion Sensors' are used in elderly care.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the principle of 'PPG' for heart rate measurement; describe how 'Motion Sensors' are used in elderly care.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure നൽകിയിരിക്കുന്നു. ഏതു result നൽകിയിരിക്കുന്നു application നൽകിയിരിക്കുന്നു. Technical terms English-ൽ എഴുതിക്കുക.

Common mistake / precaution: Motion artifacts can degrade sensor accuracy. All wearable signal processing must follow authorised noise-cancellation and motion-compensation algorithms.

2. Remote Patient Monitoring (RPM)–

RPM systems collect health data from patients at home and transmit it to healthcare providers. The architecture includes the wearable device, a mobile gateway (smartphone), and a secure cloud backend for data storage and analysis.

Study and application method

Sketch the 'RPM Architecture' showing the device, gateway, and cloud; explain the role of 'Data Visualization' in patient monitoring.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'RPM Architecture' showing the device, gateway, and cloud; explain the role of 'Data Visualization' in patient monitoring.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: രണ്ടാം concept രണ്ടാം രണ്ടാം രണ്ടാം രണ്ടാം. രണ്ടാം diagram / circuit / formula / procedure രണ്ടാം രണ്ടാം. രണ്ടാം result രണ്ടാം application രണ്ടാം രണ്ടാം രണ്ടാം രണ്ടാം രണ്ടാം. Technical terms English-ലേക്ക് രണ്ടാം രണ്ടാം രണ്ടാം.

Common mistake / precaution: RPM systems rely on stable connectivity. Do not deploy RPM for critical monitoring without authorised local-alarm and fail-safe communication mechanisms.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

IoMT and Telehealth Systems

Internet of Medical Things (IoMT): Architecture and connectivity; Telehealth and Telemedicine; Smart hospitals; Cloud-based health records; Interoperability standards (HL7, FHIR).

Approx. 16 hours

CO3

Understand · Apply

Learning goals

Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for IoMT and Telehealth Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Internet of Medical Things (IoMT)–

IoMT is a collection of medical devices and applications that connect to healthcare IT systems. It enables real-time data exchange between patients, doctors, and hospital equipment. Connectivity is achieved through Wi-Fi, BLE, and cellular networks.

Study and application method

Define 'IoMT'; list three benefits of 'IoMT' for smart hospitals; explain the role of 'Gateways' in IoMT.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'IoMT'; list three benefits of 'IoMT' for smart hospitals; explain the role of 'Gateways' in IoMT.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകുന്നു. ഏതു diagram / circuit / formula / procedure നൽകുന്നു. ഏതു result നൽകുന്നു. ഏതു application നൽകുന്നു. Technical terms English-ൽ നൽകുന്നു.

Common mistake / precaution: Network congestion can delay life-critical data. All IoMT deployments must follow authorised Quality of Service (QoS) and priority-handling protocols.

2. Telehealth and Interoperability–

Telehealth uses digital communication to provide healthcare services remotely. Interoperability standards like HL7 and FHIR (Fast Healthcare Interoperability Resources) ensure that health data can be shared securely across different systems and providers.

Study and application method

Differentiate between 'Telemedicine' and 'Telehealth' conceptually; explain the purpose of 'FHIR' in health data exchange.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'Telemedicine' and 'Telehealth' conceptually; explain the purpose of 'FHIR' in health data exchange.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യത്യാസം വ്യക്തമാക്കുക. ഡയഗ്രാം / സിരക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ ഉപയോഗിക്കുക. ഫലം വ്യക്തമായി പറയുക. ഫലം ഉപയോഗിക്കുന്ന പ്രയോഗം വ്യക്തമാക്കുക. Technical terms English-ൽ പറയുക.

Common mistake / precaution: Incompatible data formats hinder patient care. All digital health systems must follow authorised interoperability standards for seamless data integration.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Regulation, Privacy and Future Innovations

Medical device regulations: FDA, CE, CDSCO; Data privacy and security: HIPAA, GDPR; Ethics in healthcare AI; Future trends: Bio-integrated electronics, Digital twins.

Approx. 14 hours

CO4

Understand · Apply

Learning goals

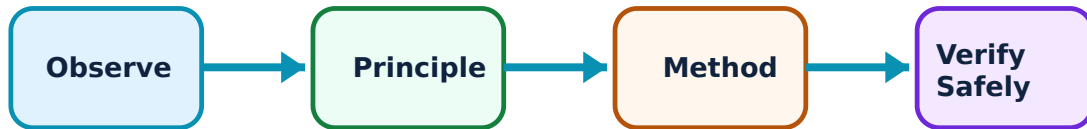
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Regulation, Privacy and Future Innovations: identify the system, state its principle, follow the correct method and verify the result safely.

1. Regulatory Compliance and Privacy–

Medical devices must meet strict regulatory standards for safety and efficacy. Data privacy is governed by laws like HIPAA (US) and GDPR (EU), which mandate secure storage and patient consent for data sharing.

Study and application method

List three 'Regulatory Bodies' for medical electronics; explain the importance of 'HIPAA' compliance in healthcare IT.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three 'Regulatory Bodies' for medical electronics; explain the importance of 'HIPAA' compliance in healthcare IT.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ diagram / circuit / formula / procedure ഈ result application ഈ technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Data breaches in healthcare can lead to severe legal and ethical consequences. All healthcare electronic systems must follow authorised encryption and access-control standards.

2. Future Innovations in Healthcare–

Future trends include bio-integrated electronics (implants), digital twins (virtual models of patients), and AI-driven predictive diagnostics. These technologies aim to provide personalized and proactive healthcare.

Study and application method

Describe the concept of a 'Digital Twin' in healthcare; list one future innovation in 'Bio-integrated Electronics'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the concept of a 'Digital Twin' in healthcare; list one future innovation in 'Bio-integrated Electronics'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈശ്വരം ഈശ്വരം ഈശ്വരം. ഈശ്വരം diagram / circuit / formula / procedure ഈശ്വരം ഈശ്വരം. ഈശ്വരം result ഈശ്വരം application ഈശ്വരം ഈശ്വരം ഈശ്വരം ഈശ്വരം. Technical terms English-ഈ ഈശ്വരം ഈശ്വരം.

Common mistake / precaution: Ethical considerations in AI diagnostics are paramount. Do not rely on AI-based health predictions without authorised clinical validation and ethical review.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Introduction to Healthcare Electronics and Wearables

Use the concept flow to revise observation, principle, method and verification.

Module 2: Wearable Sensors and Monitoring Systems

Use the concept flow to revise observation, principle, method and verification.

Module 3: IoMT and Telehealth Systems

Use the concept flow to revise observation, principle, method and verification.

Module 4: Regulation, Privacy and Future Innovations

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare RPM and Telemedicine in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a 'Wearable Sensor Network' showing multiple sensors connected to a smartphone.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the 'IoMT Data Flow' diagram from a sensor to a physician's dashboard.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the 'Battery Life' conceptually for a wearable device with a given power profile.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a healthcare electronic app or device (e.g., Apple Health, Fitbit) and note its features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Introduction to Healthcare Electronics and Wearables

Explain Healthcare Technology Evolution and state one correct method, application or precaution. –

Healthcare electronics has evolved from bulky hospital equipment to portable and wearable devices. Digitization and connectivity enable continuous monitoring outside clinical settings. Key drivers include aging populations, rising healthcare costs, and the need for preventive care.

Answer framework: List three 'Emerging Trends' in healthcare electronics; explain the benefit of 'Wearable Technology' for chronic disease management.

Important point: Wearable data is for monitoring, not self-diagnosis. Do not interpret wearable readings for clinical decisions without authorised medical consultation.

Explain Flexible and Wearable Electronics and state one correct method, application or precaution. –

Wearable devices require flexible substrates and low-power components. Smart patches use thin-film sensors for continuous glucose or ECG monitoring. Flexible batteries and antennas are essential for comfort and performance.

Answer framework: Describe the concept of 'Flexible Electronics' conceptually; list two types of 'Smart Wearables' used in healthcare.

Important point: Skin irritation and biocompatibility are major concerns. All wearable designs must follow authorised material safety and dermatological standards.

Module 2 — Wearable Sensors and Monitoring Systems

Explain Physiological Sensing in Wearables and state one correct method, application or precaution. –

PPG sensors use light to measure heart rate and SpO₂. Bio-impedance sensors measure body composition and hydration. IMUs (Inertial Measurement Units) track activity levels and detect falls. These sensors must be low-power and highly integrated.

Answer framework: Explain the principle of 'PPG' for heart rate measurement; describe how 'Motion Sensors' are used in elderly care.

Important point: Motion artifacts can degrade sensor accuracy. All wearable signal processing must follow authorised noise-cancellation and motion-compensation algorithms.

Explain Remote Patient Monitoring (RPM) and state one correct method, application

or precaution. –

RPM systems collect health data from patients at home and transmit it to healthcare providers. The architecture includes the wearable device, a mobile gateway (smartphone), and a secure cloud backend for data storage and analysis.

Answer framework: Sketch the 'RPM Architecture' showing the device, gateway, and cloud; explain the role of 'Data Visualization' in patient monitoring.

Important point: RPM systems rely on stable connectivity. Do not deploy RPM for critical monitoring without authorised local-alarm and fail-safe communication mechanisms.

Module 3 — IoMT and Telehealth Systems

Explain Internet of Medical Things (IoMT) and state one correct method, application or precaution. –

IoMT is a collection of medical devices and applications that connect to healthcare IT systems. It enables real-time data exchange between patients, doctors, and hospital equipment. Connectivity is achieved through Wi-Fi, BLE, and cellular networks.

Answer framework: Define 'IoMT'; list three benefits of 'IoMT' for smart hospitals; explain the role of 'Gateways' in IoMT.

Important point: Network congestion can delay life-critical data. All IoMT deployments must follow authorised Quality of Service (QoS) and priority-handling protocols.

Explain Telehealth and Interoperability and state one correct method, application or precaution. –

Telehealth uses digital communication to provide healthcare services remotely. Interoperability standards like HL7 and FHIR (Fast Healthcare Interoperability Resources) ensure that health data can be shared securely across different systems and providers.

Answer framework: Differentiate between 'Telemedicine' and 'Telehealth' conceptually; explain the purpose of 'FHIR' in health data exchange.

Important point: Incompatible data formats hinder patient care. All digital health systems must follow authorised interoperability standards for seamless data integration.

Module 4 — Regulation, Privacy and Future Innovations

Explain Regulatory Compliance and Privacy and state one correct method, application or precaution. –

Medical devices must meet strict regulatory standards for safety and efficacy. Data privacy is governed by laws like HIPAA (US) and GDPR (EU), which mandate secure storage and patient consent for data sharing.

Answer framework: List three 'Regulatory Bodies' for medical electronics; explain the importance of 'HIPAA' compliance in healthcare IT.

Important point: Data breaches in healthcare can lead to severe legal and ethical consequences. All healthcare electronic systems must follow authorised encryption and access-control standards.

Explain Future Innovations in Healthcare and state one correct method, application or precaution.—

Future trends include bio-integrated electronics (implants), digital twins (virtual models of patients), and AI-driven predictive diagnostics. These technologies aim to provide personalized and proactive healthcare.

Answer framework: Describe the concept of a 'Digital Twin' in healthcare; list one future innovation in 'Bio-integrated Electronics'.

Important point: Ethical considerations in AI diagnostics are paramount. Do not rely on AI-based health predictions without authorised clinical validation and ethical review.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Introduction to Healthcare Electronics and Wearables.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Wearable Sensors and Monitoring Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: IoMT and Telehealth Systems.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Regulation, Privacy and Future Innovations.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Evolution of healthcare technology; Wearable electronics: Smartwatches, Fitness trackers, Smart patches; Flexible and Stretchable electronics; Market trends and impact..
2. Develop a complete answer or practical plan for: Wearable sensors: PPG (Photoplethysmography), Bio-impedance, Temperature, Motion (IMU); Signal acquisition and processing; Remote Patient Monitoring (RPM) architecture..
3. Develop a complete answer or practical plan for: Internet of Medical Things (IoMT): Architecture and connectivity; Telehealth and Telemedicine; Smart hospitals; Cloud-based health records; Interoperability standards (HL7, FHIR)..
4. Develop a complete answer or practical plan for: Medical device regulations: FDA, CE, CDSCO; Data privacy and security: HIPAA, GDPR; Ethics in healthcare AI; Future trends: Bio-integrated electronics, Digital twins..

LAST-MILE REVISION

Quick Revision

Module 1: Introduction to Healthcare Electronics and Wearables

Evolution of healthcare technology; Wearable electronics: Smartwatches, Fitness trackers, Smart patches; Flexible and Stretchable electronics; Market trends and impact.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Wearable Sensors and Monitoring Systems

Wearable sensors: PPG (Photoplethysmography), Bio-impedance, Temperature, Motion (IMU); Signal acquisition and processing; Remote Patient Monitoring (RPM) architecture.

- Match each topic to CO2.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 3: IoMT and Telehealth Systems

Internet of Medical Things (IoMT): Architecture and connectivity; Telehealth and Telemedicine; Smart hospitals; Cloud-based health records; Interoperability standards (HL7, FHIR).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Regulation, Privacy and Future Innovations

Medical device regulations: FDA, CE, CDSCO; Data privacy and security: HIPAA, GDPR; Ethics in healthcare AI; Future trends: Bio-integrated electronics, Digital twins.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Healthcare Technology Evolution	Healthcare electronics has evolved from bulky hospital equipment to portable and wearable devices.
Flexible and Wearable Electronics	Wearable devices require flexible substrates and low-power components.
Physiological Sensing in Wearables	PPG sensors use light to measure heart rate and SpO2.
Remote Patient Monitoring (RPM)	RPM systems collect health data from patients at home and transmit it to healthcare providers.
Internet of Medical Things (IoMT)	IoMT is a collection of medical devices and applications that connect to healthcare IT systems.
Telehealth and Interoperability	Telehealth uses digital communication to provide healthcare services remotely.
Regulatory Compliance and Privacy	Medical devices must meet strict regulatory standards for safety and efficacy.
Future Innovations in Healthcare	Future trends include bio-integrated electronics (implants), digital twins (virtual models of patients), and AI-driven predictive diagnostics.

LEARNING RESOURCES

References

Recommended healthcare electronics references

1. Edward Sazonov and Michael R. Neuman, Wearable Sensors: Fundamentals, Implementation and Applications.
2. M. R. Yuce and T. Dissanayake, Wireless Body Area Networks.
3. R. S. Khandpur, Telemedicine: Technology and Applications.
4. Various, IEEE Transactions on Biomedical Engineering and Digital Health.

Approved public healthcare electronics sources

- <https://nptel.ac.in/courses/108108124>
- https://onlinecourses.nptel.ac.in/noc25_ee142/preview

These sources provide the verified study basis while official Course 6044D detail is unavailable; they do not replace official clinical designs, authorised medical device certifications, certified data privacy standards or authorised strategic healthcare plans.